

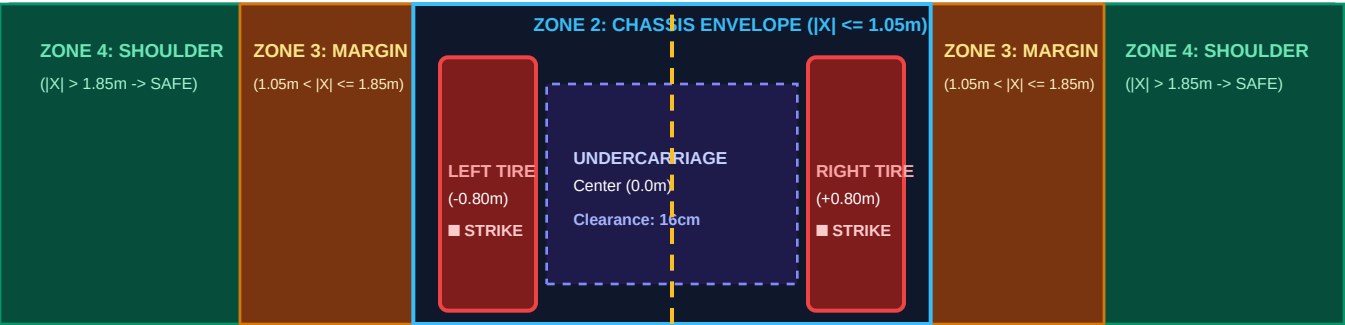
SAFAR: Lane & Ego Driving Corridor System

Geometric Lateral Reasoning, Wheel Track Collision Protection & Trajectory Lookahead

1. Executive Purpose: Eliminating Phantom Braking

In complex urban and highway driving, obstacles and surface hazards exist constantly in adjacent lanes, on sidewalks, and along road shoulders. A naive safety system that brakes for every detected object will cause dangerous false alarms. The **SAFAR Lane & Driving Corridor System** provides continuous lateral geometric boundary evaluation to determine if an anomaly physically intersects the vehicle's driving path.

2. Multi-Zone Geometric Corridor Architecture



3. The Four Lateral Corridor Zones

Zone	Coordinates (X)	Strike Target	Lateral Multiplier	Safety Action & Control Logic
Zone 1: Wheel Tracks	Left: $-0.80m \pm 0.125m$ Right: $+0.80m \pm 0.125m$	Tires / Rims / Steering Knuckle	1.00 (Max)	Direct wheel strike. Applies active braking for deep potholes/craters.
Zone 2: Chassis Envelope	$ X \leq 1.05m$ (Total: 2.10m)	Undercarriage / Oil Pan	0.90 (if $D > 14cm$) 0.40 (if $D \leq 12cm$)	Evaluates depth vs 16cm ground clearance. Safe shallow ruts pass over.
Zone 3: Lane Margin	$1.05m < X \leq 1.85m$	Adjacent Lane / Grazing	0.50 (Moderate)	Tagged as POSSIBLE_INTERSECTION. Speed moderated without hard braking.
Zone 4: Road Shoulder	$ X > 1.85m$ to 10.0m	Sidewalk / Opposing Lane	0.05 (Ignored)	Tagged as PATH_CLEAR. 100% ignored. Manual driving uninterrupted.

4. Dynamic Curved Path & Steering Projection

When the vehicle turns, the corridor projects dynamically along the steering radius rather than remaining straight:

Curved Path Formulation: $X_{\text{effective}}(d) = X_{\text{measured}} - 0.5 * (v^2 / R) * \tan(\text{delta_steering}) * (d / v)$

- **Steering Left:** Shifts the future corridor leftward. Obstacles straight ahead safely exit the corridor.
- **Steering Right:** Shifts the corridor rightward, detecting hazards inside the curve.

5. Trajectory Lookahead & Crossing Threat Prediction

For crossing pedestrians and merging two-wheelers, SAFAR projects lateral motion: $X_{\text{target}}(t) = X_0 + V_{\text{lat}} * t$. If a pedestrian currently in Zone 4 (sidewalk) will enter Zone 2 (chassis envelope) at the exact arrival time ($t = \text{TTC}$), SAFAR flags a **Pre-Corridor Crossing Threat** and initiates early deceleration.

6. Implementation Map

- **safar/pothole/path.py**: Computes wheel strike overlap and undercarriage ground clearance checks.
- **safar/pothole/config.py**: Defines `VEHICLE_HALF_WIDTH_M = 1.05m` and `LANE_HALF_WIDTH_M = 1.75m`.
- **Prediction/SAFARTrajectoryPredictor.h**: 60 Hz UE5 C++ trajectory predictor for curved paths and cut-in traffic.